

Senior Thesis

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```
rhfs <- read.csv("RHFS_2024_Public_Variables.csv")

# Turning Categorical Variables into Factors
rhfs <- rhfs |>
  mutate(
    across(
      c(NUMCAT_R, OUTCOME, PROPCON, PROPTOWN,
        COMPLEX, NUMBLD_R, YRNEWBLG_R, YRPROPACQ_R, APPRWAIVER,
        CASHDWN, HOWACQ, HOWMVAL, OFFSTPARKAHS, UNITSTAT,
        ADAGENCY, ADAGENT, ADNEWSPAPER, ADONLINE, ADOTHONLINE,
        ADVERTISE, ADWEBSITE, MLS, MNGMNT, NUMOTHPROP,
        NUMOTHUNITS, OWNENT, OWNOTHPROP, COMMSpace, CONCSS,
        ICABLE, IELEC, IGAS, IPARKING, IPOOL,
        ISEWER, ITRASH, IWATER, IWIFI, LEASLNG,
        LNGTRMPRV, NORNTAPPFEE, PAYAPP, PAYCASH, PAYCHK,
        PAYCREDIT, PAYONLINE, PAYOTH, PAYTHIRD, RCONTROLAHS,
        RECOMMREN, RNTAPPFEWAIV, SECDEPAMT, SENIOR, WEEKLNG,
        OPTAX_R, PROPANS, ACCEPT, BENEOTH, ECONDEVLP,
        ELDERLY, GOVBELOW, GOVGRANT, HISTORIC, LOWINAREA,
        LOWINTEN, OTHRCV, RSUBSIDY, STLOWINC, YRMORT_R,
        DEBT, DEBT1, GREENMORT, LOCTYP, MORTGET,
        PMTFREQ, REGPAY, RPIPMI, RPIPRIN, RPIPRPINS,
        RPIPRPTAX, WHYNOMRT, SAMPSTRATA, MORTAMT_R, OUTBAL_R, RNTAPPFEE_R,
        YRPROPREHAB_R),
      as.factor
    )
  )

# Renaming some factor levels for later convenience
rhfs <- rhfs |>
  mutate(
    NUMCAT_R = fct_recode(
      NUMCAT_R,
      "1 Unit" = "0",
      "2-4 Units" = "1",
      "5-24 Units" = "2",
      "25-49 Units" = "3",
      "50+ Units" = "4"
    ),
    OUTCOME = fct_recode(
      OUTCOME,
      "Interview Compl; self response" = "1",
```

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    "Assisted living limited interview; self response" = "3",
    "Conversion from partial to full interview" = "5",
    "Interview Compl; field interview" = "201",
    "Assisted living limited interview; field interview" = "203",
    "Sufficient partial; field interview" = "204",
    "Military limited field interview" = "206"
  )
)

```

```

# Loads the data frame with srvyr
df <- rhfs |>
  select(-REPWGT0) |> # REPWGT0 is equivalent to WEIGHT
  as_survey_rep(
    repweights = starts_with("REPWGT"),
    weights = WEIGHT,
    type = "successive-difference",
    combined_weights = TRUE,
    # Decided to set to T to follow SDR as outlined by Ash (to match what was
    # done in Census's Methodology)
    mse = TRUE
  )

# Sanity check for if the data was successfully loaded with srvyr
# using Table 5 of Census's RHFS Methodology
df |>
  summarize(
    properties = survey_total(vartype = "se") # Applies weights and calculates se of properties
  ) |>
  mutate( # Dividing by 1000 to match Table 5's scale
    properties = properties / 1000,
    properties_se = properties_se / 1000,
    moe = properties_se * 1.645 # 90% margin of error
  )

```

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## # A tibble: 1 x 3
##   properties properties_se   moe
##   <dbl>         <dbl> <dbl>
## 1    18965.         165.  271.

```

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# Makes View(rhfs) much more productive.
rhfs <- rhfs |>
  select(-starts_with("REPWGT"), -WEIGHT,
    -SURVYR, -CONTROLPUF, # is always 2024, ID isn't useful in data analysis
    # J flag variables mark values that were manually reviewed; note that
    # they may become relevant if outliers are under scrutiny.
    -starts_with("J"),
    # A continuous variable coded as a factor without documentation; may
    # be useful for pure prediction
    -PCTFEWAIV_R
  )

# NAs in the dataset
missingvalues <- function(df) {

```

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NAs <- sum(sapply(df, \(x) sum(x == -8))) # Not applicable
NRs <- sum(sapply(df, \(x) sum(x == -9))) # Not reported

tibble(
  NAs = NAs, # Not applicable
  NRs = NRs, # Not reported
  perc.missing = (NAs + NRs) / prod(dim(df)) # % of missing values in the df
)
}
missingvalues(rhfs)

```

```

## # A tibble: 1 x 3
##       NAs    NRs perc.missing
##   <int> <int>      <dbl>
## 1 171343 145654      0.405

```

```

# Missing values by interview type
rhfs |>
  summarize(
    # pick(everything()) extracts a data frame of all non-grouping columns
    missingvalues(pick(everything())),
    .by = OUTCOME
  )

```

```

##                               OUTCOME   NAs   NRs perc.missing
## 1          Interview Compl; field interview 97854 73886    0.3966648
## 2      Sufficient partial; field interview 30024 36928    0.5211083
## 3          Interview Compl; self response 38033 21317    0.3261276
## 4      Conversion from partial to full interview 3448  7818    0.4572240
## 5      Military limited field interview    300   836    0.7171717
## 6 Assisted living limited interview; field interview 1286 3761    0.7169034
## 7 Assisted living limited interview; self response   398 1108    0.7130682

```

```

# Handling known Trivially Missing values
# Certain values in the columns below can only exist if REGPAY is 1, so -9 is
# wrongly assigned to them.
trivially.missing.regpay <- rhfs |>
  mutate(id = row_number()) |>
  filter(
    REGPAY == -9 &
    # In given observation, if any of these columns contain a value other than
    # -8 or -9 return True.
    if_any(c(PMTAMT_R, PMTFREQ, RPIPRIN, RPIPRPINS, RPIPMI, RPIPRPTAX), \(x) !x %in% c(-8, -9))
  ) |>
  select(id, REGPAY, PMTAMT_R, PMTFREQ, RPIPRIN, RPIPRPINS, RPIPMI, RPIPRPTAX) |>
  pull(id)

rhfs$REGPAY[trivially.missing.regpay] <- 1

# Evidence for MAR (missing at random) in MRKTVAL_R
df |>
  mutate(

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mrktNA = MRKTVAL_R == -9
) |>
summarize(
  survey_mean(mrktNA),
  .by = NUMCAT_R
)

```

```

## # A tibble: 5 x 3
##   NUMCAT_R      coef  '_se'
##   <fct>      <dbl>  <dbl>
## 1 1 Unit      0.0923 0.0263
## 2 2-4 Units   0.0716 0.0224
## 3 5-24 Units  0.154  0.0224
## 4 25-49 Units 0.335  0.0547
## 5 50+ Units   0.418  0.0163

```

```

df |>
filter(MRKTVAL_R != -9) |>
summarize(
  mean_mrktval = survey_mean(MRKTVAL_R),
  median_mrktval = survey_median(MRKTVAL_R),
  .by = NUMCAT_R
)

```

```

## # A tibble: 5 x 5
##   NUMCAT_R      mean_mrktval mean_mrktval_se median_mrktval median_mrktval_se
##   <fct>      <dbl>      <dbl>      <dbl>      <dbl>
## 1 1 Unit      397050.      42218.      250000      33154.
## 2 2-4 Units   569582.      33304.      425000      50491.
## 3 5-24 Units  2178309.     118540.     2000000     469164.
## 4 25-49 Units  5000928.     333704.     4000000     224403.
## 5 50+ Units  33895580.     2865340.     20000000     4987831.

```